

Date	3 JULY 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Team Involvement in Demonstration

Introduction

The **Credit Card Approval Prediction using Machine Learning** project was developed and demonstrated as an individual academic project. Although the project template includes a section for team involvement, all activities related to the planning, development, implementation, testing, documentation, and demonstration were independently carried out by a single developer. Every phase of the project was completed with careful planning and execution to ensure that the final solution met the project objectives and demonstrated the complete machine learning development lifecycle.

Individual Contributions

The project began with identifying the problem statement and understanding the challenges involved in detecting fraudulent credit card transactions. This was followed by collecting the dataset, performing exploratory data analysis, preprocessing the data, handling missing values, feature scaling, and preparing the dataset for machine learning model development.

Multiple machine learning algorithms, including Logistic Regression, Decision Tree, and Random Forest, were implemented and evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Based on the comparative analysis, the Random Forest Classifier was selected as the final prediction model due to its superior performance.

A Flask-based web application was then developed to integrate the trained machine learning model with an interactive HTML and CSS user interface. The application allows users to enter transaction details and receive instant predictions indicating whether a transaction is genuine or fraudulent. Additional work included integrating the preprocessing modules, loading the trained model, validating user inputs, and displaying prediction results through a user-friendly interface.

Demonstration Activities

During the project demonstration, every stage of the application was presented in a systematic manner. The presentation included an explanation of the project objectives, dataset, preprocessing techniques, machine learning model development, evaluation results, web application architecture, testing procedures, and future enhancement plans. Live demonstrations were performed to show the prediction workflow, starting from entering transaction details to displaying the final prediction result.

The project documentation, GitHub repository, application screenshots, architectural diagrams, data flow diagrams, empathy map, brainstorming activities, and other supporting documents were also prepared independently and presented during the demonstration to provide a complete understanding of the project.

Learning Outcome

Developing and demonstrating the project individually provided valuable practical experience in machine learning, data preprocessing, web application development using Flask, project

documentation, software testing, and version control using GitHub. The project also improved problem-solving abilities and strengthened the understanding of integrating machine learning models into real-world web applications.

Conclusion

The successful completion and demonstration of the Credit Card Approval Prediction project reflect the ability to independently manage every stage of an end-to-end machine learning application. From project planning and implementation to testing, documentation, deployment, and final presentation, all responsibilities were carried out individually. The demonstration effectively showcased the functionality, performance, and practical significance of the developed system while highlighting its potential for future improvements and real-world applications.